STUDENT PERFORMANCE: EDUCATIONAL EQUITY & INSIGHTS

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GOOGLE COLAB LINK :

https://colab.research.google.com/drive/1x8oKalnBRAA1Xvauyj5ZR_WRScrGiJXF?usp=sharing

Introduction

1. This empirical report investigates the structural factors influencing student academic achievement across key subjects: Mathematics, Reading, and Writing. Utilizing a standardized student performance dataset (N = 1,000), this study examines how external demographic features—specifically parental level of education, lunch socio-economic indicator, and completion of test preparation courses—correlate with academic outcome variations.

Primary Dataset Size

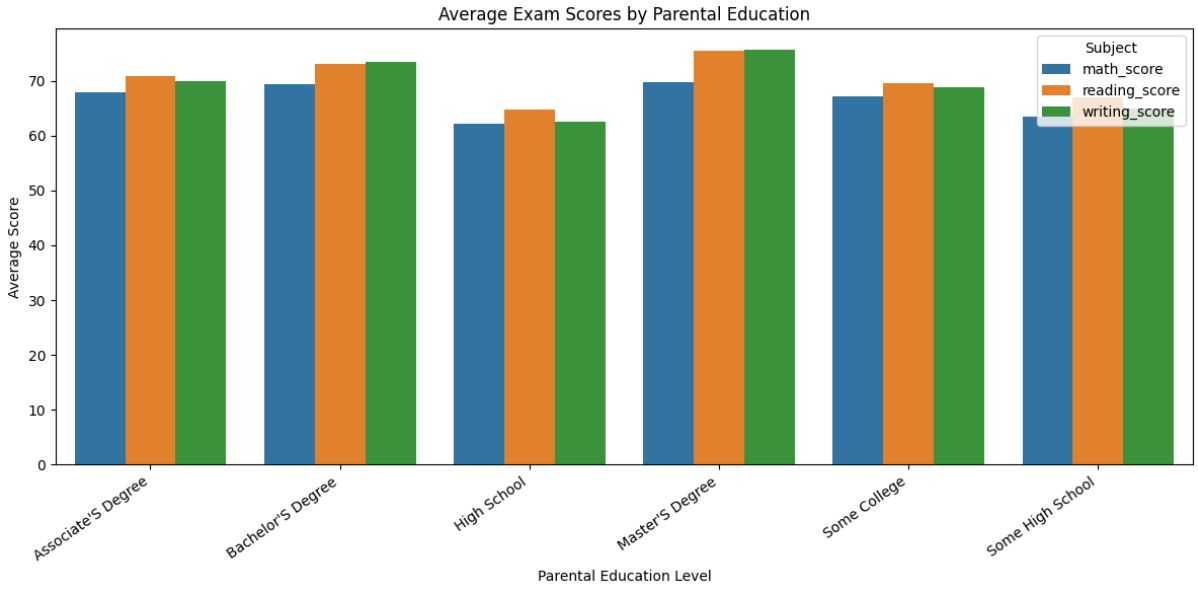
1,000 Students Core Performance Metrics

Math, Reading, Writing

The core objective is to uncover equity gaps in learning outcomes and provide evidence-based recommendations for targeting support mechanisms, resource allocation, and curriculum planning without making simplistic causal assertions.

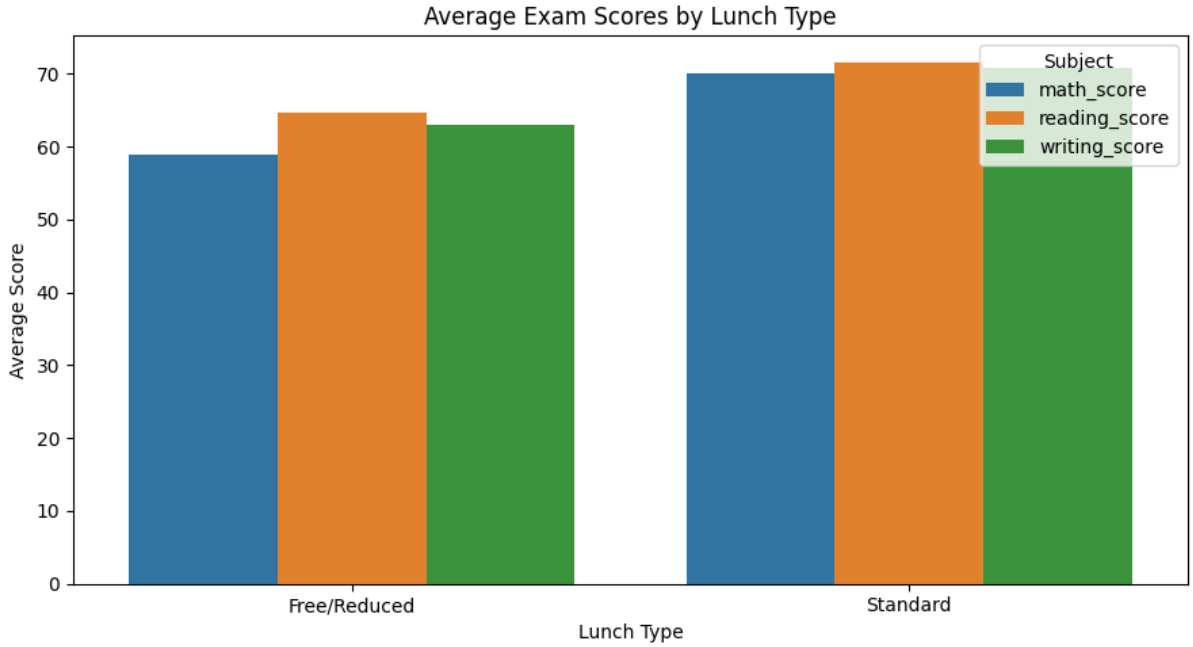
2. Analysis of Parental Education

Parental education level is widely documented as an environmental determinant in student outcome variances. The data reveals a consistent positive trend: students whose parents hold post-secondary degrees (Master's or Bachelor's) systematically achieve higher mean scores across all three subjects compared to those whose parents completed high school or some high school.



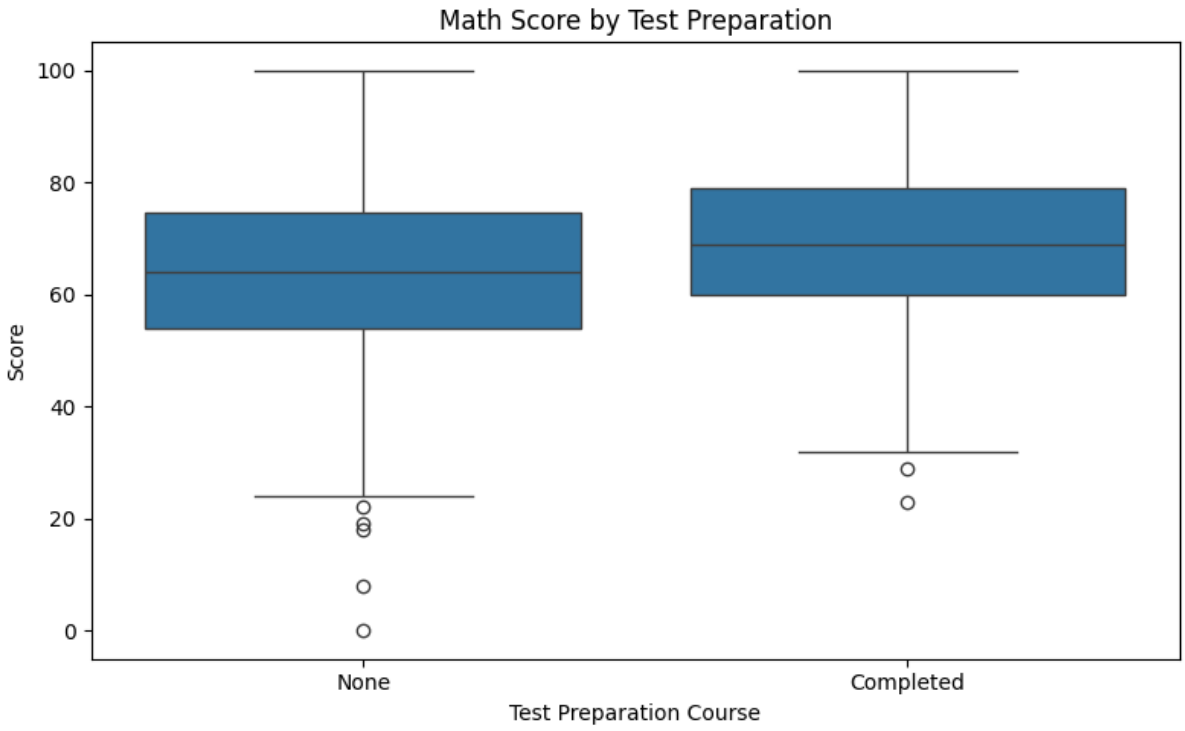
3. Analysis of Lunch Type

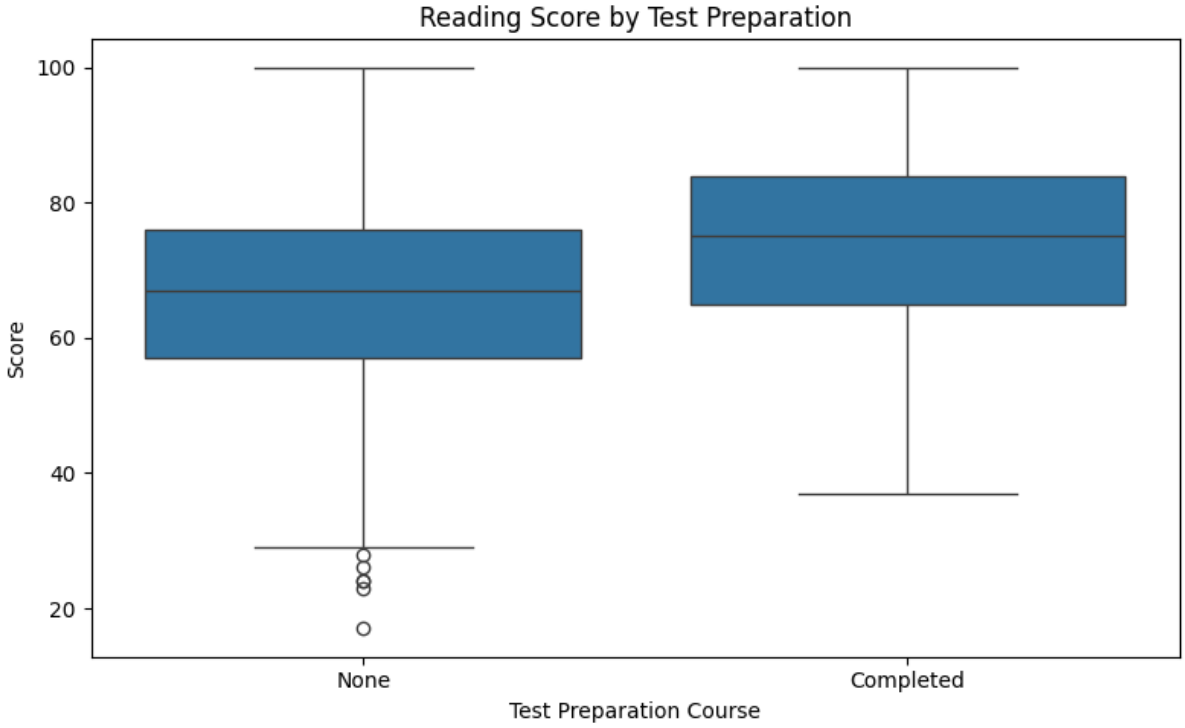
Lunch subscription type (Standard vs. Free/Reduced) serves as a proxy indicator for household socioeconomic background. A substantial baseline discrepancy exists between the two categories across all academic domains. Students receiving standard lunch demonstrate higher mean scores in Math (70.0), Reading (71.6), and Writing (70.8), whereas students on free or reduced lunch average lower outcomes (Math: 58.9, Reading: 64.6, Writing: 63.0).

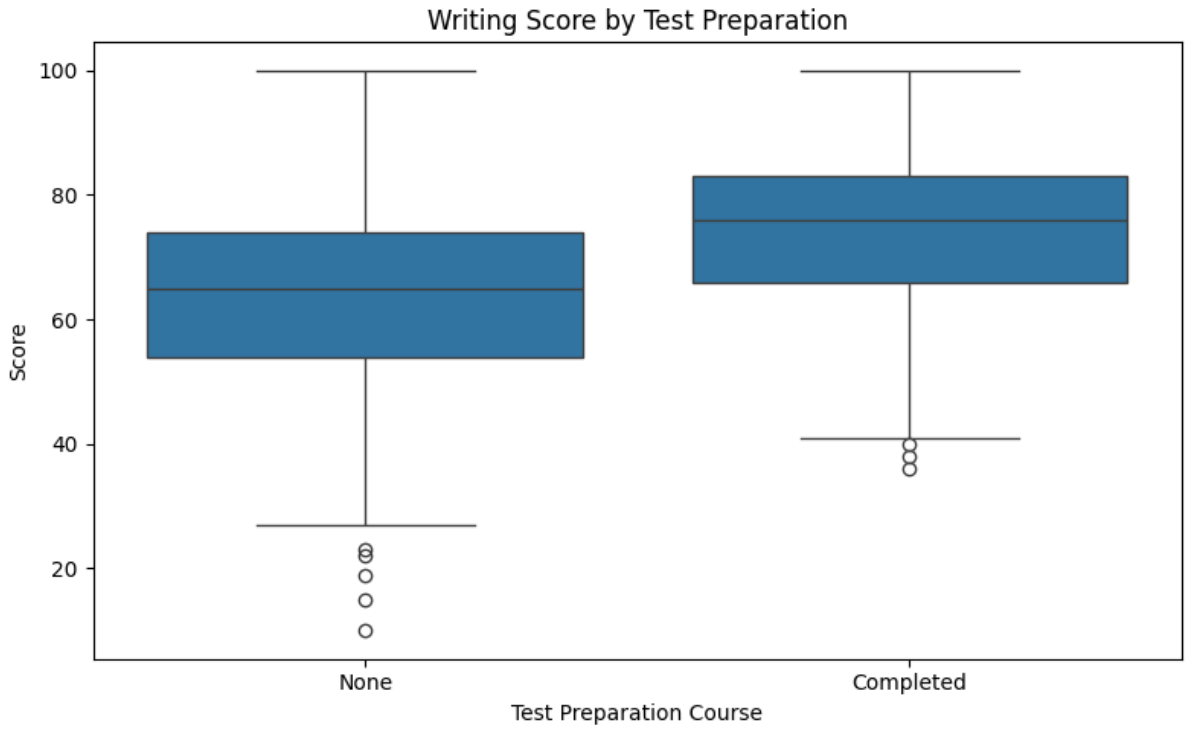


4. Test Preparation Analysis

Participation in targeted test preparation courses yields a clear upward shift in performance distributions. A comparative analysis of median values and score variance shows that course completion reduces low-end performance outliers and raises median scores in Reading and Writing by nearly 8–10 points, with significant improvements in Math as well.

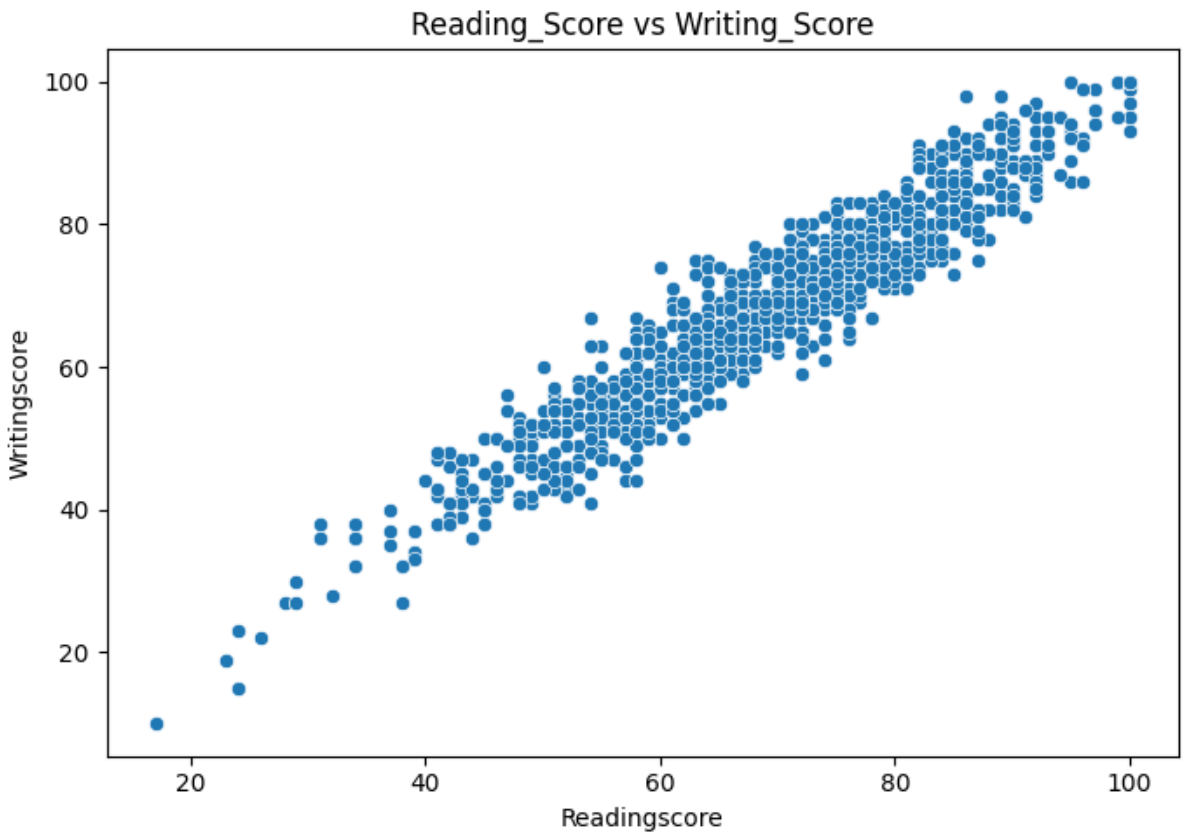

Math Score by Test Preparation
Score
100
80
60
40
20
0
None
Completed
Test Preparation Course

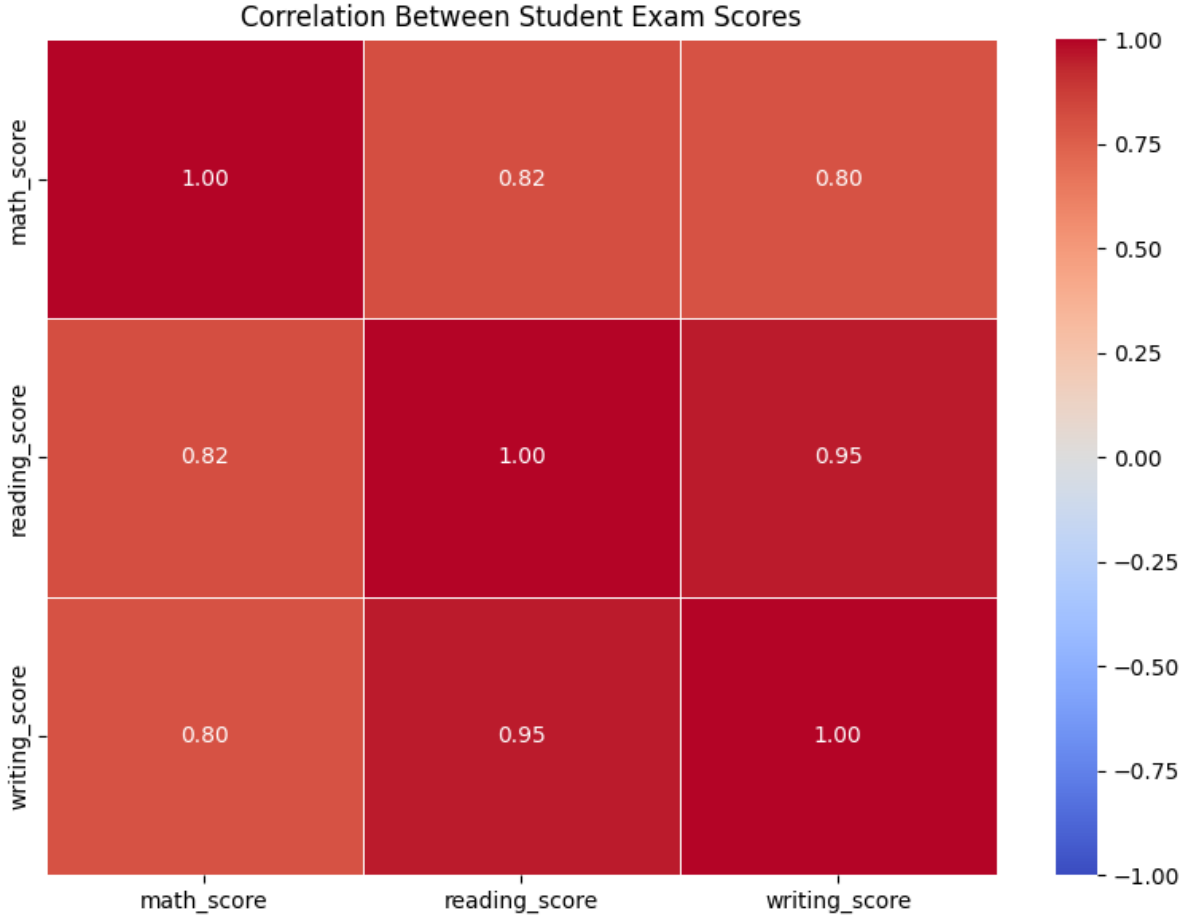

Reading Score by Test Preparation
Score
100
80
60
40
20
None
Completed
Test Preparation Course



5. Subject Correlation Analysis

Evaluation of subject-to-subject correlation coefficients reveals an exceptionally strong linear relationship between Reading and Writing scores ($r = 0.95$), suggesting shared underlying verbal comprehension and literacy skill sets. Correlation between Math and verbal subjects remains strong ($r \approx 0.80$), confirming overall academic alignment across disciplines.


Reading_Score vs Writing_Score
Writingscore
Readingscore
100
80
60
40
20
20
40
60
80
100


Correlation Between Student Exam Scores
math_score
reading_score
writing_score
1.00
0.82
0.80
0.82
1.00
0.95
0.80
0.95
1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

Educational Equity Recommendations

Based on empirical findings across parental background, lunch status, and course intervention, the following institutional steps are recommended:

• Targeted Academic Support: Allocate supplementary tutoring and instructional resources to students exhibiting lower baseline performance, specifically within under resourced socioeconomic demographics.

• Universal Access to Test Preparation: Expand access to subsidized or school-integrated test preparation courses to eliminate cost barriers for lower-income households.

• Cross-Subject Literacy Initiatives: Leverage high reading writing correlations by integrating writing-intensive exercises directly into mathematics and science modules.

• Continuous Demographic Outcome Monitoring: Establish periodic disaggregated performance tracking to evaluate equity gaps without creating stigmatizing interventions.

• Evidence-Based Evaluation: Assess intermediate intervention pilots using additional longitudinal data prior to executing large-scale structural policy shifts.

Limitations

While the analysis identifies notable empirical patterns, several critical analytical limitations must be acknowledged:

• Non-Causal Relationship: Observed group differences and correlation coefficients establish statistical association rather than direct causation. Unobserved confounding variables (e.g., household stability, access to private educational media, regional funding discrepancies) may drive score variances.

• Proxy Variables: Subsidized lunch status serves as an indirect indicator for family income, which may not capture full household wealth or nuanced socioeconomic conditions.

• Omitted Factors: The dataset does not account for qualitative parameters such as teacher effectiveness, student attendance records, individual learning accommodations, or psychological well-being.

8. Summary

In conclusion, educational outcomes remain tied to structural socioeconomic markers, but targeted interventions such as structured test preparation demonstrate clear effectiveness in mitigating score disparities. Institutional efforts should focus on democratic access to academic prep programs and cross disciplinary support.